北京郵電大學

Queen Mary University of London

For examiners' use only

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Total	

EBU5477 A

Joint Programme Examinations 2023/24

EBU5477 Embedded Systems

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

Allowed: (approved) electronic calculators
NOT allowed: Any Electronic devices including electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Matthew Tang, Dr Jin Zhang

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Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) Imagine that you are an embedded system designer in an electronic consumer product company. You are given a context diagram of the design of a Bluetooth speaker, as shown in **Figure 1**. Study the diagram carefully and answer the following questions.

[15 marks]

i) Based on the information provided in **Figure 1**, sketch a hardware block diagram for the speaker. Your diagram should focus on the physical objects in the embedded system. For each connection of the blocks, identify a possible hardware interface (e.g. GPIO, ADC, etc.).

(5 marks)

ii) Sketch an organigram that illustrates the hierarchy of the embedded software for the speaker. You should make sure that the information is consistent with your answer in **(i)**.

(5 marks)

iii) One of your colleagues suggests that using recycled aluminium for the case of the speaker can improve the durability of the product by 50% with an increase of the cost of the whole product for 25%. **In your opinion**, what would be the impact to the environment if this suggestion is accepted? Hence, state whether you agree or disagree to the suggestion.

(5 marks)

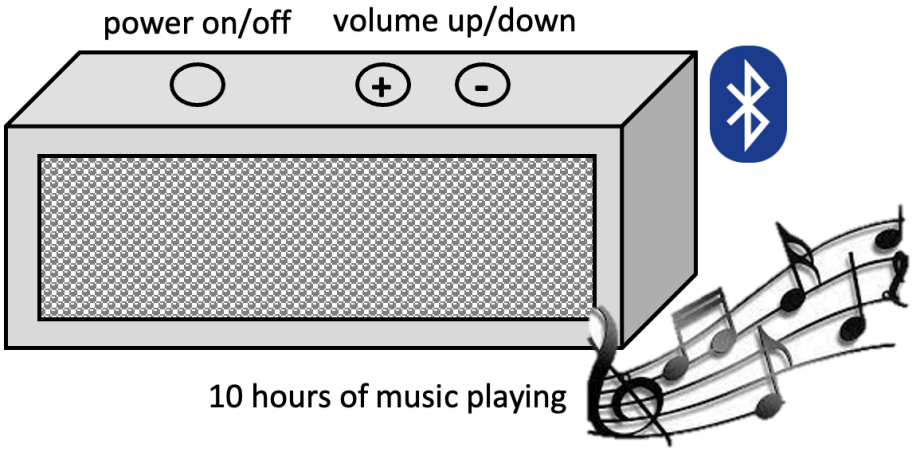


Figure 1: Context Diagram of a Bluetooth Speaker

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	15 marks

b) Two ARM Cortex-M4 instructions are shown below:

```
MOV r0, r1
ADD r2, r0, r1
```

It is known that r0 = 4, r1 = 5, r2 = 6 before the instructions were executed.

[5 marks]

i) What are the final values in the registers r0, r1 and r2 respectively after the execution?

(3 marks)

ii) What would be the total size (in bytes) of the two instructions if both are known to be very common?

(2 marks)

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	5 marks

c) An ARM Cortex-M4 microcontroller maps all its physical SRAM to the memory addresses between 0x20000000 and 0x2001FFFF. Calculate the size (in bytes) of the physical SRAM.

[5 marks]

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	5 marks

Question marking: $\frac{}{15} + \frac{}{5} + \frac{}{5} = \frac{}{25}$

Question 2

a) The snippet of a main program for an ARM Cortex-M4 processor is shown in **Figure 2**.

[10 marks]

```
int func1(int a, int b){
    int x, y;
    x = a * a + b * b;
    y = sqrt(x); // square root function from math.h
    return y;
}
void main(void){
    …
    d = func1(3, 6);
    …
}
```

Figure 2: Code snippet of an Cortex-M4 main program

i) According to the Procedure Call Standard for the ARM Architecture (AAPCS), what registers should be used to pass arguments *a* and *b* from **main()** to **func1()?**

(2 marks)

ii) Integers *x* and *y* inside **func1()** are called *automatic variables*. Explain how memory space would be allocated for these two variables.

(3 marks)

iii) What must be done in the beginning of **func1()** (i.e. prolog) such that the program could return to **main()** eventually? And why?

(3 marks)

iv) How would the answer *y* from **func1()** be returned back to **main()**?

(2 marks)

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	10 marks

b) A snippet of a C program for an ARM Cortex-M4 processor is shown in **Figure 3**. It works with the LCD display and the UART hardware using the standard drivers as in our laboratory exercises.

Explain briefly the meaning of the C statements at

(i) line 2; (ii) line 4, (iii) line 7, and (iv) lines 08-11.

[5 marks]

```
01: void main_test(void){
02:     uart_init(9600);
03:     uart_enable();
04:     lcd_init();
05:
06:     while (1) {
07:         char c = uart_rx();
08:         if (c == '0')
09:             lcd_clear();
10:         else
11:             lcd_put_char(c);
12:     }
13:}
```

Figure 3: Code snippets of a Cortex-M4 program for LCD and UART

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	5 marks

c) Consider a system built on a 48 MHz Cortex-M4 processor with a main loop and only a single interrupt which occurs at a 30 kHz frequency. Assume that the interrupt service routine (ISR) takes 600 clock cycles to execute, and there is a total of 12 clock cycles of response and return time overhead.

[10 marks]

i) What percentage of the processor's time (utilisation) is spent in the ISR and the overhead? So what is the percentage of the processor's time left for the main loop to execute?

(4 marks)

ii) If the main loop requires 55 ms of computation to execute one iteration, calculate the maximum main loop update rate (in Hz)?

(2 marks)

iii) State two benefits of interrupt over polling

(4 marks)

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		10 marks

Question marking: $\frac{}{10} + \frac{}{5} + \frac{}{10} = \frac{}{25}$

Question 3

a) Two GPIO ports are used to demonstrate the addition of two bits. GPIO port A (GPIO_A) is used to configure two GPIO pins (Pin 1: SW1 and Pin 2: SW2) to read the status of two different switches. GPIO port B (GPIO_B) is used to configure two GPIO pins (Pin 1: LED1 and Pin 2: LED2) are used to control two LEDS based on the status of the switches, as in **Table 1**.

[15 marks]

Table 1: SW and LED relationship

	SW1	SW2	LED1	LED2
Initial state	**0**	**0**	**0**	**0**
	0	1	0	1
	1	0	0	1
	1	1	1	0

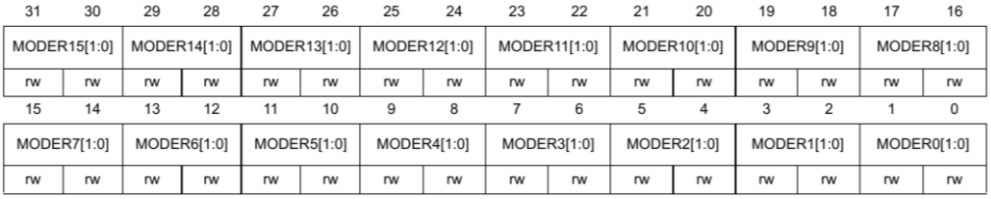


Figure 4: MODER register

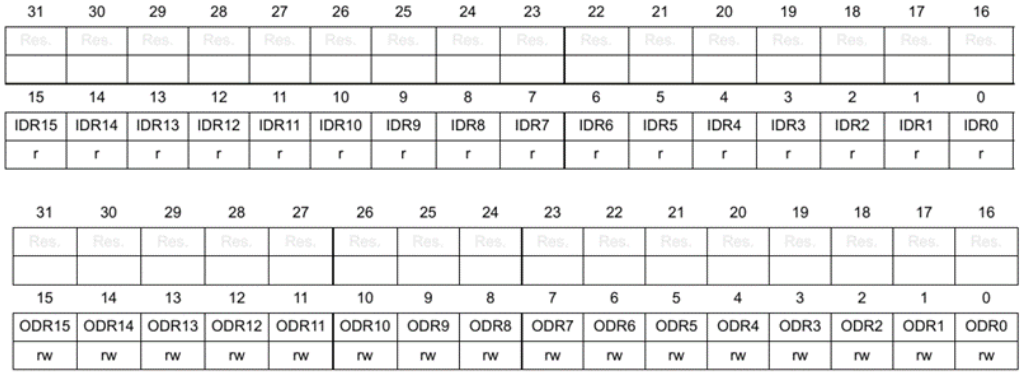


Figure 5: IDR and ODR registers

i) Write **one possible value** of registers MODER_A and MODER_B (see **Figure 4**).

- 00: Input (reset state)
- 01: General purpose output mode
- 10: Alternate function mode
- 11: Analog mode

Hint: It is enough to show the first word of MODER (bit 15 to bit 0), e.g., for MODER with 0 everywhere, you can write 0000

(4 marks)

ii) Write **one possible value** of registers IDR_A, IDR_B, ODR_A, and ODR_B when SW1=1 after the initial state (see **Figure 5**)

(6 marks)

iii) Write **one possible value** of registers IDR_A, IDR_B, ODR_A, and ODR_B when SW2=1 after **(ii)** above.

(5 marks)

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b) This question relates to asynchronous serial communication as shown in **Figure 6**.

[4 marks]

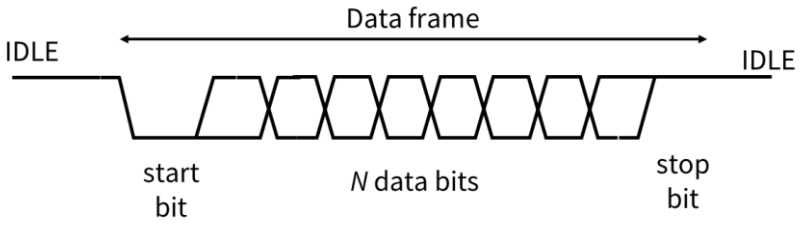


Figure 6: UART Data Frame

The following data frame is transmitted: 1010101X, where the last bit (X) is a parity bit. Answer the following questions:

i) What is the value of N in this case?

(1 mark)

ii) What is the value of X based on the received decisions in **Table 2**. Is it Odd or Even parity?

(3 marks)

Table 2: Parity Decision Table

Received data frame	**Decision**
10101110	No Error
00101011	Error

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	4 marks

c) This question relates to the USART baud rate setting for an $f_{PCLF} = 12MHz$ and a desired Baud rate of 460.8Kbps. Note that *OVER8*=0 indicates oversampling by 16 and *OVER8*=1 indicates oversampling by 8. The relation between f_{PCLF} and the actual baud rate is shown in **Figure 7** and depends on the setting of *OVER8* and *USARTDIV*. The value of *USARTDIV* can be obtained based on *DIV_Mantissa* and *DIV_Fraction* also in **Figure 7**.

[6 marks]

$$T_x\backslash R_x\ (baud) = \frac{f_{PCLF}}{8 \times (2 - OVER8) \times USARTDIV}$$

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Reserved															
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
DIV_Mantissa[11:0]												DIV_Fraction[3:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Figure 7: Baud rate equation and the DIV register

i) What should be the value of *USARTDIV* and *DIV_Fraction* for oversampling by 16?

(4 marks)

ii) What would be the error between the desired Baud rate and the actual one?

(2 marks)

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	6 marks

Question marking: $\frac{}{15} + \frac{}{4} + \frac{}{6} = \frac{}{25}$

Question 4

a) Suppose you are going to flash an LED on and off periodically as shown in **Figure 8**, using delay by software only or hardware timer. The processor clock frequency is 84 MHz. Answer the following questions.

[12 marks]

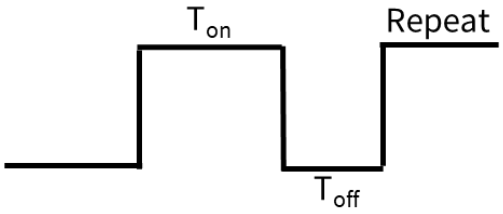


Figure 8: Periodic pulse for LED flashing

i) A student wrote the following loop for a time delay. Do you think it is a reasonable method? Explain briefly

```
for (i = 0; i < 1000; i++);
```

(4 marks)

ii) If the SysTick timer is used to create the pulses, calculate the initial value for the SysTick timer if you want to count for 50 μs.

(4 marks)

iii) If T_{on}=150 μs, T_{off}=100 μs, based on **(ii)**, suggest how to use SysTick timer to get a delay for the on/off states.

(4 marks)

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	12 marks

b) A Cortex M4 microcontroller (MCU) connects to 3 LCD displays using the I^2C bus. The I^2C slave addresses for the displays are 0x28, 0x29, and 0x2A respectively. Answer the following questions.

[13 marks]

i) Can you connect to more LCD displays, and why?

(3 marks)

ii) Draw a diagram to show the electrical signals (with high and low levels) on the I^2C buses when the MCU tries to address and write to the LCD of the slave address 0x28 successfully.

(7 marks)

iii) When two (or more) masters (M1 & M2) begin a transfer at the same time, what technique can be used to decide which master can get the I^2C bus access?

(3 marks)

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